



IDX G9 BIOLOGY S STUDY GUIDE ISSUE S2

FINALS

By Amy and Mya, Edited by Isabella

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This guide is a combined guide. Due to the final range including content from before the monthlies, we have included content from the previous issue in this guide for your convenience.

18.1 Finding Order in Diversity

Assigning Scientific Names

- ❖ Common names can be confusing because they vary among **languages** from place to place, and different species may **share a common name**
 - In the 18th century, scientists recognized this confusion, so they agreed to assign **Latin or Greek** names to each species
- ❖ Names could be **long and difficult** to standardize these names as different scientists focused on **different traits** --> didn't clear up much confusion

Dichotomous Key

- ❖ Used to **identify organisms**
 - A series of paired statements or questions that describe alternative characteristics of an organism. A paired statement usually describes the presence or absence of a certain **visible characteristic or structure**

Binomial Nomenclature

- ❖ 1730s, Swedish botanist [Carolus Linnaeus](#)
- ❖ Two-part scientific name written in *italic*
 - First word (genus) begins with a capital letter and the second word (species) is lowercased
 - ◆ Ex. *Ursus Maritimus*
- ❖ Species name often describes of important trait or the organism's habitat
- ❖ Writing Standards:
 - The first name is the genus name, the second name is the species name
 - The 1st letter of the genus name is capitalized
 - The species name is lowercased
 - Italics are used if the name is typed or printed
 - The name is underlined if handwritten
 - Genus name can be abbreviated to the initial letter after used in text

Classifying Organisms into Larger Groups

- ❖ **Systematics:** Science of naming and grouping organisms
 - Goal is to organize living things into groups (taxa, single: taxon) that have [biological meaning](#)

The Linnean Classification System

- ❖ Original classification system expanded to include seven hierarchical taxa
 - Kingdom, Phylum, Class, Order, Family, Genus, Species
 - King Philip Came Over For Green Soup
 - Kindly Put Clothes On Fat Grey Squirrels
- ❖ Classified according to [visible similarities and differences](#)
- ❖ But appearances can have [mistakes](#) (ex. Look different but DNA alike) and it's hard to identify which characteristics are [considered important](#)
 - Modern systematics apply [Darwin's ideas of evolution](#) to classification and look beyond to ask questions about evolutionary relationships

- ◆ Assign species to larger group in ways that reflect how closely member of groups related

18.2 Modern Evolutionary Classification

Evolution: descent with modification

Phylogeny: the evolutionary history

Phylogenetic systematics: evolutionary classification, to group species into categories that reflect lines of evolutionary descent, rather than overall similarities and differences.

Clades

- ❖ consist of an ancestor organism and all its descendants (and nothing else)
- ❖ must be **monophyletic group**: includes all species that are descended from a common ancestor

Cladistics: a method of classifying species of organisms into groups called clades

Paraphyletic group: consists of an ancestral species and some, but not all, of the descendants.

Clades and Traditional Taxonomic Groups (Taxa)

- ❖ Cladistic analysis shows that many traditional taxonomic groups do form valid clades.
 - Eg: Linnaean class Mammalia, for example, corresponds to clade Mammalia.
- ❖ In other cases, however, traditional groups do not form valid clades.
 - Eg: Linnaean class Reptilia, which does not include birds, is therefore not a valid clade and not corresponds to clade Reptilia
- ❖ Mammals have the unique homologous characteristic of producing milk. They form a clade
- ❖ Likewise, birds share the common characteristic of feathers. They also form a clade.
- ❖ Reptiles, as a group, consist of the crocodilians, lizards and snakes, tortoises and turtles and tuatara. However, they are not a clade. One of them is actually more closely related to birds.

- ❖ Instead, **birds, dinosaurs, crocodiles, and all descendants living or extinct**) of their **most recent** common ancestor form a **clade**.

Cladogram

- ❖ a tree diagram to link clades together
- ❖ links groups of organisms by showing how evolutionary lines, or lineages, branched at from common ancestors.

Building Cladogram

- ❖ The bottom, or "root", of the cladogram represents the common ancestor shared by all organisms on the cladogram
- ❖ **Nodes**
 - the branching points on a cladogram
 - denote **a speciation event** when a common ancestor splits into two, or more, species
- ❖ **Branching patterns indicate degrees of relatedness among organisms.**

Derived Characters

- ❖ A trait that arose in the most recent common ancestor of a particular lineage clade and was passed along to its descendants

Losing Trait

- ❖ The absence of a trait is not used as a derived character, because distantly related groups of organisms can lose the same character
- ❖ Example: both whales and snakes have lost the tetrapod character of four limbs, but they are not very closely related. Snakes are members of the clade Reptilia, while whales are members of the clade Mammalia.

18.3 Building the Tree of Life

Classification of Living Things						
DOMAIN	Bacteria	Archaea	Eukarya			
KINGDOM	Eubacteria	Archaeobacteria	"Protista"	Fungi	Plantae	Animalia
CELL TYPE	Prokaryote	Prokaryote	Eukaryote	Eukaryote	Eukaryote	Eukaryote
CELL STRUCTURES	Cell walls with peptidoglycan	Cell walls without peptidoglycan	Cell walls of cellulose in some; some have chloroplasts	Cell walls of chitin	Cell walls of cellulose; chloroplasts	No cell walls or chloroplasts
NUMBER OF CELLS	Unicellular	Unicellular	Most unicellular; some colonial; some multicellular	Most multicellular; some unicellular	Most multicellular; some green algae unicellular	Multicellular
MODE OF NUTRITION	Autotroph or heterotroph	Autotroph or heterotroph	Autotroph or heterotroph	Heterotroph	Autotroph	Heterotroph
EXAMPLES	<i>Streptococcus</i> , <i>Escherichia coli</i>	Methanogens, halophiles	<i>Amoeba</i> , <i>Paramecium</i> , slime molds, giant kelp	Mushrooms, yeasts	Mosses, ferns, flowering plants	Sponges, worms, insects, fishes, mammals

*King Proista is not a true clade! (paraphyletic)

Changing ideas about Domains

- All organism are classified into three domains
- Based on comparing base sequences of **ribosomal RNA (rRNA)** which evolves slowly

A domain is a larger, more inclusive category than a kingdom

Changing ideas about Kingdoms

- Previously prokaryotes were placed in one kingdom (**monera**) and eukaryotes in the remaining four
- Recent evidence from genetic studies of **rRNA** has shown that "prokaryotes/monerans" are far more diverse.

19.1 The Fossil Record

Fossils—the preserved remains or traces of ancient life

Types of Fossils

Fossils found in **rock** include **petrified** fossils, **molds and casts**, and **trace fossils**.

Petrified fossils

- fossils in which minerals replace all or part of an organism

Mold

- a hollow area in sediment while a cast is a copy of the shape an organism (or part of it)

Trace fossils

- provide evidence of the activities of ancient organisms such as footprint, burrow

other fossils form when the remains of organisms are preserved in substances [such as tar, amber, or ice](#) (original organic matter)

How do fossils form?

- Specific conditions are needed for [fossilization](#)
- Fossils are usually found in [sedimentary rock](#)—rock that is made of hardened sediment.
 1. Small particles (sand, silt, clay, or lime muds) settle to the bottom of water body and bury dead organisms.
 2. The remains are buried deeper as sediment continue to build up over time.
 3. Water pressure gradually compresses the lower layers and turns the sediments into rock
 4. The preserved remains may later be discovered and studied.

From the fossil record, paleontologists learn about

- Ancient organisms'
 - ❖ structure
 - ❖ environment
 - ❖ the ways in which they lived
 - ❖ when they lived

Dating Earth's History

Relative dating

- Places rock layers and their fossils into a **temporal sequence** (whether a fossil is older or younger than other fossils)

Index fossils

- can provide the **relative** age of a rock layer
- existed **only** during **specific** spans of time occurred in large geographic areas
- eg: trilobites

Radiometric dating (direct/absolute dating)

- compares the amount of radioactive to nonreactive isotopes in a sample to determine its age.

Radioactive isotopes decay into nonradioactive isotopes at a steady rate.

Half-life

- the time required for half of the radioactive atoms in a sample to decay.

Geologic Time Scale

- Based on both **relative** and **absolute** dating
- Relative ——— rock layers and index fossils
- Absolute ——— radiometric dating techniques
- Geologists used rock boundaries to determine where one division of geologic time ended
- A timeline of Earth' s history is called the geologic time scale ——— **eons, eras, and periods**
- All geologic time before the Cambrian is simply called **Precambrian time** (covering about 90% of Earth's history)
 - ❖ Eons ———divided into eras
 - ❖ Eras ———divided into periods
 - ❖ Periods
 - ◆ most commonly used unit of time on the geologic time scale

- ◆ associated with rock systems

Life on a Changing Planet

Physical forces

- building mountains, opening coastlines, changing climates, geological forces (altering habitats of living organisms repeatedly throughout Earth history), collision of comets and asteroids, etc.
- the theory of plate tectonics explains how solid continental "plates" move slowly above Earth's molten core —a process called **continental drift**

Biological forces

- actions of living organisms over time have changed conditions in the land, water, and atmosphere
 - ❖ The first photosynthetic organisms began absorbing carbon dioxide and releasing large amounts of oxygen
 - ❖ Earth cooled down when CO₂ levels dropped
 - ❖ Iron content of the oceans fell, as iron ions reacted with oxygen to form insoluble compounds that settled to the ocean floor

19.2 Patterns and processes of evolution

Microevolution

- the change in allele frequencies that occurs over time within a population.

Macroevolution

- the grand transformations in anatomy, phylogeny, ecology, and behavior, which usually take place in clades larger than a single species
 - ❖ Speciation of larger clades
 - ❖ Extinction of larger clades

If the **rate of speciation** in a clade is **equal** to or greater than the rate of extinction, the clade will continue to exist. If the rate of extinction in a clade is greater than the rate of speciation, the clade will eventually become extinct.

Pace of Extinction

Background extinction

- extinct because of the slow but steady process of natural selection

Mass extinction

- Extinct over a relatively short period of time (in geological time scale)
- Entire ecosystems vanish
- whole food web collapse
- Natural selection can't compensate quickly.
- Combination of multiple factors: volcanic eruptions, moving continents, changing sea levels, etc. eg: giant asteroid crashed at the end of Cretaceous Period

Evidence shows that evolution has often proceeded at different rates for different organisms at different times.

Pace of Speciation



■ Gradualism

- Darwin
- Evolution occurred by a long sequence of continuous change.
- Species slowly change through a series of intermediate forms
- For example the change in size and hoof of the modern horse
- incomplete fossil record to explain the lack of inter- mediates

■ Punctuated Equilibrium

- Stephen J. Gould and Niles Eldredge
- Long periods of relative stability in a species are punctuated by periods of rapid evolution.
- No long sequence of intermediate forms.
- For example, horseshoe crabs (*The species has not changed for a very long time.*)
- Much more common in organisms with short generation times

Rapid evolution

- **After a small population becomes isolated from the main population**
- **when a small group of organisms migrates to a new environment**
- **after mass extinctions, which open many ecological niches, creating new opportunities for those organisms that survive.**

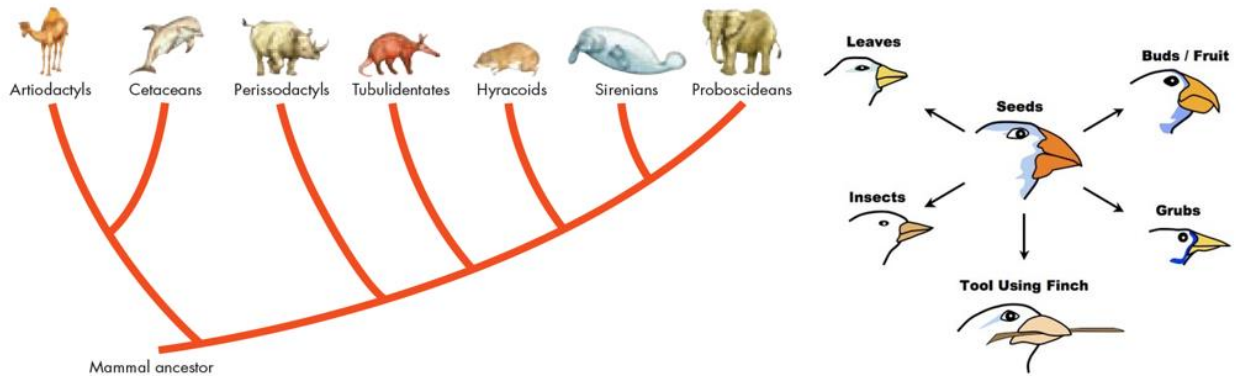
Patterns of Evolution

Coevolution

- **Two species evolve in response to changes in each other over time.**
- **Eg1: flowers and pollinators**
- **Eg2: plants and herbivorous insects**
 - **Plants evolved bad-tasting or poisonous compounds**
 - **Insects evolved variants that could alter, inactivate, eliminate poisons**

Adaptive Radiation

- A single species evolves over a relatively short time into several different forms
- Species display variations on the ancestral structure
- Species occupy different ecological niches
- Eg: Galápagos finches, Hawaiian honeycreepers



Convergent Evolution

- Groups of organisms evolve in different places or at different times, but in similar environments.
- Organisms face similar selection pressures
- Produces similar adaptations
- For example, emus, rheas, and ostriches



FIGURE 19-12 Convergent Evolution Mammals that feed on ants and termites evolved independently five times. Although each species is unique, each has evolved powerful front claws, a long hairless snout, and a tongue covered with sticky saliva. These adaptations are useful for hunting and eating insects.



19.3 Earth's Early History

Life's Origin

- ❖ Formation of earth: Cosmic debris collided with one another
- ❖ 4.6 billion years ago --> Volcano eruptions, comets and asteroids
- ❖ 4.2 billion years ago --> Solid rocks form, water condensed enough to fall as rain

❖ Earth's early atmosphere contain **little or no oxygen**

- Principally composed of carbon dioxide, water vapor, and nitrogen with lesser amounts of carbon monoxide, hydrogen sulfide, and hydrogen cyanide

First organic molecules

❖ 1953: Stanley Miller and Harold Urey's experiment

- Stimulate early earth in flasks and tubes (oceans, atmosphere, lightning, rain)
- Suggest how mixture of organic compounds could've risen from **simpler compounds**
- Composition of early atmosphere incorrect but production of organic compounds correct

Formation of microspheres

❖ Appeared during Archean Eon

❖ Large organic molecules form tiny bubbles called **proteinoid microspheres**

Microspheres

- **NOT cells** but have characteristics of living systems
 - ◆ Selectively permeable membranes for water to pass
 - ◆ Simple ways of storing and releasing energy

Evolution of RNA and DNA

❖ Remember cells controlled by information stored in DNA, which is transcribed to RNA and then translated into proteins

- But how did they evolve?

❖ RNA hypothesis

- RNA could have formed from simpler molecules
- Existed **before DNA**
- Could lead to DNA-directed protein synthesis
- Still being tested

Production of free oxygen

❖ **First life forms of prokaryotes**

- **3.5 billion years ago**
- **Evolved in absence of oxygen**

❖ **Photosynthetic bacteria-producing oxygen**

- **2.2 billion years ago**
- **Oxygen mix with iron in oceans --> Iron oxide (rust) which sunk to ocean floor**
- **Oxygen accumulate in atmosphere --> Ozone layer**
 - ◆ **New organisms evolve using oxygen for respiration while some early life forms became extinct**

Origin of eukaryotic cells

❖ **Prokaryotic cells**

- **No nucleus**

❖ **Eukaryotic cells**

- **Have nucleus**
- **Complex organelles**
- **Have mitochondria**
- **Plants and algae have chloroplasts**

❖ **Endosymbiotic theory**

- **Proposes that a [symbiotic relationship](#) evolved over time, between primitive eukaryotic cells and the prokaryotic cells within them**
 - ◆ **Prokaryotic cells entered those ancestral eukaryotes**
 - ◆ **Imagine your food becoming your organs**

Sexual reproduction and multicellularity

- ❖ **Development of sexual reproduction sped up evolutionary change because it increases genetic variation**
- ❖ **Asexual reproduction**

- Duplicate genetic material and pass it on to daughter cells

- Basically **restricts mutations** in DNA

- ❖ Sexual reproduction

- Meiosis and fertilization shuffle genes generating **genetic diversity**

- Increase populations' adapting to new or changing environmental conditions

- ❖ Multicellularity

- Evolved after cellular respiration

- Undergo series of **adaptive radiations** that result in great diversity